

BOOBALAN A

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Professional Summary

Data Scientist with end-to-end experience across the full data science lifecycle—from raw telematics data ingestion and EDA to feature engineering, machine learning model development, and production deployment on GCP. Experienced in geospatial intelligence, predictive analytics, and automated MLOps pipeline development within the commercial vehicle domain. Comfortable owning problems end-to-end, from SQL queries and data analysis to stakeholder-facing dashboards and productionized solutions.

Experience

VE Commercial Vehicles (VECV) | *Data Scientist I*

Mar 2025 – Present

Geospatial & Telematics Analytics | GCP | Python | MLOps

- **Reducing False Positives in DSMS (Ongoing) — YOLO:** Developing a YOLO-based computer-vision pipeline to classify and filter false-positive alerts generated by the Driver Safety Monitoring System (DSMS) using **image-based analysis**; applying object detection on captured frames to validate alert triggers, improving alert precision and reducing false alarm rates for fleet managers.
- **Geospatial Clustering Engine (HAC + Haversine):** Designed and deployed a production geospatial model that identifies high-frequency vehicle stoppage zones at **500 m precision** across a fleet of commercial trucks, enabling the operations team to reduce idle-time incidents — directly supporting route optimisation decisions for field logistics.
- **Stoppage Mining Module:** Developed a BigQuery-based stoppage mining pipeline applying 5 telemetry validity checks, Haversine-based distance correction, and POI-radius mapping to generate validated stoppage events; automated MySQL/CSV export with run-tracking and failure-alert logging.
- **MLOps Pipeline — 99.9% Uptime, Zero Credential Exposure:** Architected an end-to-end CI/CD pipeline on GCP (Cloud Build, Cloud Scheduler, Secret Manager) with OIDC keyless auth, replacing a manual cron-based process; **eliminated all hardcoded secrets** and maintained **99.9% uptime** across 15-minute automated data-processing jobs, cutting deployment cycle from hours to minutes.
- **Fuel Analytics Algorithm Optimisation:** Partnered with the Fuel Analytics team to audit and retune telematics algorithms for fuel consumption calculation; improvements reduced signal-noise ratio in sensor data, increasing model accuracy and surfacing actionable fuel-efficiency insights for fleet managers.
- **Fuel Sensor Classification (Reed vs. Float) — 96% Accuracy:** Built and deployed an end-to-end pipeline on GCP to automatically classify vehicle fuel sensors using telematics data from BigQuery; applied break-point detection and angle-based horizontal-segment analysis on fuel time-series, aggregated predictions across **100+ fert codes** with rule-based confidence gating (**99%+**) — replacing 100% manual inspection and running autonomously every month in production.

Advart Technologies | *Data Analyst Intern*

Dec 2024 – Feb 2025

E-commerce Analytics | Python | Predictive Modelling | Dashboard Engineering

- **Sell-Out Forecasting Model:** Built and shipped a time-series predictive model (Python / Scikit-learn) that forecast daily sell-out and monthly GMV turnover for an e-commerce client; replaced gut-feel inventory decisions with a **data-driven restock trigger**, reducing overstock risk.
- **Self-Service Analytics Dashboard:** Delivered an interactive Python-based sales dashboard that condensed 5+ manual weekly reports into a single real-time view, cutting analyst reporting time and giving the product team on-demand visibility into conversion and revenue metrics.

Rabwin Industries | *Data Analyst Intern*

Jul 2024 – Dec 2024

Power BI | Automated Reporting | EDA | Multi-source Data Integration

- **Reporting Efficiency +30%:** Automated a previously manual Power BI reporting workflow by integrating multi-system sales data into a single refreshable pipeline — **cut report generation time by 30%** and freed the sales team from weekly manual data pulls.
- **Cross-System EDA:** Led exploratory analysis across three disparate data sources (ERP, CRM, warehouse), uncovering sales pattern anomalies that were previously invisible due to siloed reporting.

Technical Skills

Languages	Python (Pandas, NumPy, Scikit-learn, SciPy, Matplotlib, Seaborn), SQL
ML / Data Science	Machine Learning, Deep Learning, Predictive Modelling, Time-Series Forecasting, Geospatial Analysis (HAC, Haversine, DBSCAN), EDA, Feature Engineering
MLOps & Cloud	GCP (Cloud Build, Cloud Scheduler, Secret Manager, Compute Engine, IAM, BigQuery), CI/CD Pipelines, OIDC Keyless Auth, Docker (basics)
BI & Visualisation	Power BI, Interactive Python Dashboards (Plotly / Dash), Tableau (familiar)
Domain	Telematics, Fleet Analytics, Commercial Vehicle Data, IoT Sensor Data, E-commerce Analytics
Soft Skills	Cross-functional collaboration, Stakeholder communication, End-to-end problem ownership

Education

M.Sc. Data Science — Amrita Vishwa Vidyapeetham (Amrita University) 2020 – 2025

GPA: 7.5 / 10 | Coursework: Statistical Learning, Deep Learning, Big Data Systems, Machine Learning, Image processing.